

Install Manual

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1. DeskPRO Download Installation Manual

Installing and maintaining DeskPRO Download requires basic server administration skills. If your organization does not have someone with the correct skills, we suggest you use [DeskPRO Cloud](#), our hosted service.

If you run into an installation issue, check the [Troubleshooting](#) section of this manual and search our [Knowledgebase](#).

We also offer an [installation service](#) for an additional fee.

Note that the most recent version of DeskPRO is always hosted at <https://www.deskpro.com/downloads/deskpro.zip>. We recommend you install the newest version as we are always adding features and improvements.

1.1. Requirements

DeskPRO is web-based software written in PHP. It is designed to be independent of the underlying operation system and webserver. It will run on any recent version of Windows (Server 2003 or later), Linux/Unix or Mac OS X.

Installing DeskPRO requires the following:

A webserver: any webserver that works well with PHP can run DeskPRO: Apache, nginx and Microsoft IIS are all supported.

PHP: minimum supported version is v5.3.2, but we recommend always using the latest version.

DeskPRO will need access to the command line version of PHP (called php/php-cli on Linux or php-win.exe on Windows). This is generally installed by default as part of the PHP installation process.

PHP extensions that you will need to install and enable are:

- PDO with the MySQL driver installed
- mbstring or iconv
- json_encode
- session
- ctype
- dom
- tokenizer
- an image manipulation extension: GD or Imagick or Gmagick
- OpenSSL (recommended)
- IMAP (if you want to use an IMAP email account)
- LDAP (if you want to use LDAP/Active Directory Authentication)

Depending on your platform/PHP installation method, many of these will be automatically enabled.

If you are using a version of PHP before 5.5, you should install an **opcode cache** such as [Alternative PHP Cache](#) or [Windows Cache Extension](#). Versions above 5.5 come with a built-in opcode cache (OPCache).

MySQL database: The minimum supported version is 5.0; we recommend using the latest 5.x.x. version.

cron (on a Unix-based OS) or **Scheduled Tasks** (on Windows) to run regular processes such as sending email

An email server (optional): DeskPRO can send email via an SMTP server, using the PHP mail() function or via a Gmail/Google Apps account. It can receive email from a POP3, IMAP or Exchange server or a Gmail/Google Apps account.

Hardware: DeskPRO runs well on modest hardware, and can easily run from a standard virtual machine or even virtual hosting account. If you are expecting a very busy helpdesk, the limiting factor for speed is generally the database, which can be helped by increasing the amount of RAM and using fast disks for storage. You can run all the required software on one machine; however, you may wish to separate the web server from the database server.

Third-party web hosting requirements

You should be able to install DeskPRO on most web hosting providers. Some web hosting services have limitations which are incompatible with DeskPRO.

- 1 You must have access to the **CLI version of PHP**, not the CGI/FCGI version. If your webhost hasn't installed the CLI binary, you will see DeskPRO errors saying "This script must only be run from the CLI." You can confirm what version is running by connecting with SSH and running:

```
/usr/bin/php -v
```

You should get a result like 'PHP 5.3.13 (cli)'. If it says 'php-cgi', you will need to request access the CLI binary.

- 2 Your web hosting provider must meet DeskPRO's requirements for PHP version & extensions and mySQL version (see above).
- 2 Setting up the cron job from a web-based interface (such as cPanel) will mean you unable to see error messages. We recommend setting up cron using SSH so you can see any errors.
- 3 Some web hosting providers limit the frequency of scheduled tasks. You should be able to set a cron job or Windows Scheduled Task to run once per minute to install DeskPRO.

1.2. Upgrading from previous DeskPRO versions

If you are using a much older version of DeskPRO (v3 or before) and want to update to the latest version, see this article on our Knowledgebase: <https://support.deskpro.com/kb/articles/366>.

This only applies if you have not updated your DeskPRO installation in several years.

1.3. Installation overview

Note that once you get past the basic steps of setting up the webserver and getting it to serve DeskPRO, the DeskPRO installation wizard will alert you to any configuration issues.

The installation process varies depending on your chosen platform, but the general steps (to install from scratch) are:

- 1 Install the webserver and enable PHP support and URL rewriting.
- 2 Install PHP, with the PHP extensions listed in [Requirements](#).
- 3 Install MySQL, and create a 'deskpro' user account.
- 4 Remove the default page (e.g. `index.html`) from your web server's base directory, and unzip `deskpro.zip` into it.
 - copy the resulting `config.new.php` file as `config.php`
 - set a cron job or Scheduled Task to run `cron.php` once per minute
 - edit `config.php` to add your email address and MySQL database credentials
- 5 Ensure the web server is running and go to the URL (or IP address) where it is serving DeskPRO.
- 6 The installation wizard will take you through the steps required to complete installation. It can advise you on how to fix common configuration problems and missed steps.
- 7 Once installation is complete, you can log on to DeskPRO with the account details you gave during installation. Start by going to the admin interface at `/admin/`.

2. Installing on Ubuntu

This section describes how to quickly install DeskPRO on Ubuntu/Apache from scratch, for example on a virtual machine.

Note that depending on your Ubuntu version, the Apache document root may either be `/var/www` or `/var/www/html`. If there is no `/var/www/html` folder, use `/var/www`.

2.1. Install server software (Ubuntu)

Run these commands:

```
apt-get update

apt-get -y install wget unzip mlocate

apt-get -y install apache2

apt-get install -y php5 php5-gd php5-mysql php5-curl php5-cli php5-cgi
php5-dev php5-ldap php5-gd php5-json php-apc libapache2-mod-php5

apt-get install mysql-server mysql-client
```

You will be asked to set a root password for MySQL. Remember this password.

If you are using a version of PHP before 5.5, you should also run:

```
apt-get install php5-apc
```

to install the APC opcode cache.

If you plan to use an IMAP email account with DeskPRO, install and enable the PHP IMAP extension:

```
apt-get install php5-imap
php5enmod imap5
```

2.2. Update Apache config (Ubuntu)

Run `a2enmod rewrite`

Edit the `default.conf` or `000-default.conf` file in `/etc/apache2/sites-available/`. Remember that you may need to replace `/var/www/html` with `/var/www`.

- If there is a section like this:

```
<Directory /var/www/html>
...
</Directory>
```

change `AllowOverride None` to `AllowOverride All`.

- If the file has no `<Directory /var/www/html>` section,

add:

```
<Directory /var/www/html>
AllowOverride All
</Directory>
```

below the line `DocumentRoot /var/www/html`.

Restart Apache:

```
service apache2 restart
```

2.3. Install DeskPRO (Ubuntu)

Run these commands from the Apache document root folder (`/var/www` or `/var/www/html`).

```
wget http://www.deskpro.com/downloads/deskpro.zip -O deskpro.zip
```

(download a ZIP of the latest version of DeskPRO)

```
rm -rf index.html
```

(deletes the default web page)

```
unzip deskpro.zip
```

(extract the zipped DeskPRO files)

```
rm -rf deskpro.zip
```

(delete the ZIP file)

```
cp config.new.php config.php
```

(create the config.php file with default settings)

2.4. Set cron job (Ubuntu)

Set a cron job to run your PHP CLI executable using the cron.php file in the Apache document root folder as an argument:

```
sed -i "1j* * * * * root /usr/bin/php  
/var/www/html/cron.php" /etc/crontab
```

or

```
sed -i "1j* * * * * root /usr/bin/php  
/var/www/cron.php" /etc/crontab
```

2.5. Edit config.php (Ubuntu)

Edit `config.php` to enter the MySQL password and your email address (for error alerts).

2.6. Complete final steps (Ubuntu)

Your DeskPRO helpdesk is now installed. You should open DeskPRO in your web browser at the relevant URL (or IP address) for the server and follow the final steps.

For production use, you should change ownership of the installed files so they are owned by the web server user or by a dedicated DeskPRO user account:

e.g. `chown -R www-data /var/www/html`

You should also [create a non-root mySQL user](#) with all privileges on the 'deskpro' database, and edit config.php accordingly.

3. Installing on CentOS/Red Hat 7

This article describes a quick installation of DeskPRO on CentOS/Red Hat.

3.1. Install server software (CentOS)

1

Install relevant software by running the following commands.

```
yum -y install wget unzip
```

```
yum -y install httpd
```

```
yum -y install php php-mysql php-gd php-cli php-ldap php-pecl-apc php-pdo php-com
```

```
yum -y install mariadb-server mariadb
```

Note that mariadb is a drop-in MySQL replacement used in newer versions of CentOS/Redhat. Apart from the installation/setup commands, it works the same way as MySQL for the purposes of using DeskPRO.

1

Set up database:

```
systemctl start mariadb
```

```
/usr/bin/mysql_secure_installation
```

You will be prompted for the database password; just press enter as it is not set yet.

Set a root password for MySQL; do so and remember this password. For the following steps, enter y (for Yes).

Ensure that the mySQL service starts on reboot using:

```
systemctl enable mariadb
```

3.2. Update Apache config (CentOS)

Edit the `/etc/httpd/conf/httpd.conf` file.

Find the section that begins

```
<Directory "/var/www/html">
```

and change `AllowOverride None` to `AllowOverride All`

Note that there are other `AllowOverride` statements in different parts of the file. You must make sure to edit the right statement.

Make sure Apache starts:

```
systemctl start apache  
systemctl enable apache
```

3.3. Install DeskPRO (CentOS)

Download the latest version of DeskPRO:

```
wget http://www.deskpro.com/downloads/deskpro.zip -O deskpro.zip
```

(download a ZIP of the latest version of DeskPRO)

```
unzip deskpro.zip
```

(extract the zipped DeskPRO files)

```
rm -rf deskpro.zip
```

(delete the ZIP file)

```
cp config.new.php config.php
```

(create the config.php file with default settings)

```
mv /var/www/html/data /var/www/data
```

(moves the DeskPRO data directory outside of the web root so it cannot be accessed via the web)

edit /var/www/html/config.php:

- 1 Set DP_DATABASE_PASSWORD to the MySQL root password.
- 2 Set DP_TECHNICAL_EMAIL to your email address (for error alerts).
- 3 Enter the information requested in the PATHS section of config.php.

You can find the path of an installed binary using `which` e.g.

```
which php
```

- 4 Set a cron job to run config.php once per minute:

```
sed -i "1i* * * * root /usr/bin/php  
/var/www/html/cron.php" /etc/crontab
```

3.4. Complete final steps (CentOS)

Your DeskPRO helpdesk is now installed. You should open DeskPRO in your web browser at the relevant URL (or IP address) for the server and follow the final steps.

For production use, you should change ownership of the installed files so they are owned by the web server user or by a dedicated DeskPRO user account:

e.g. `chown -R apache /var/www/html`

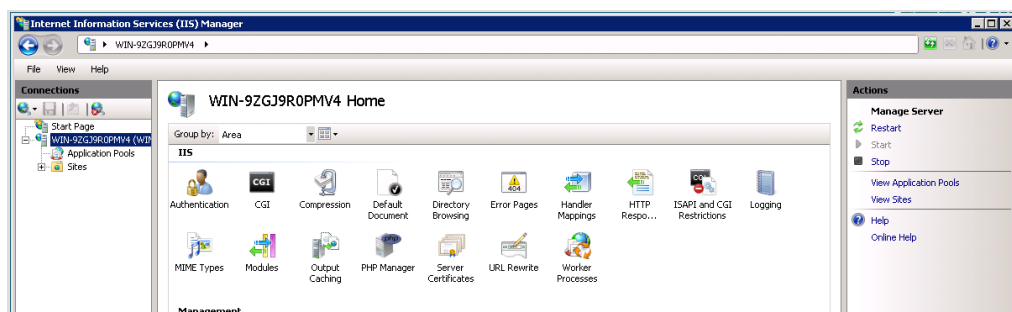
You should also [create a non-root mySQL user](#) with all privileges on the 'deskpro' database, and edit config.php accordingly.

4. Installing on Windows Server 2008/IIS 7

This section describes how to quickly install DeskPRO on Windows Server 2008.

4.1. Install server software (2008)

- 1 In Server Manager, go to **Roles** and click **Add Roles**.
 - Select **Web Server (IIS)** and under **Application Development**, select **CGI**.
 - Agree to installing any requirements.
- 2 Install the Microsoft Web Platform Installer from the Microsoft website.
- 3 Using the WPI, install:
 - PHP (v 5.5 recommended)
 - URL Rewrite 2.0
 - These should install automatically; if not, install them too:
 - IIS 7.0 Fast CGI
 - PHP Manager for IIS
- 4 Open IIS Manager (under **Administrative tools** in the Start menu).



Open the PHP Manager panel, fix any configuration issues that PHP Manager flags up, and click **Check phpinfo()** to display information a

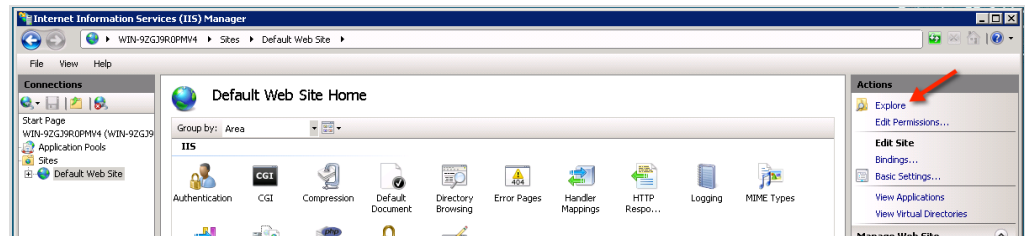
- 5 Install the latest MySQL server; an easy way to do this is using [MySQL Installer](#), select **Server Only** (despite the name, this also installs the command line client) and set the configuration type as **Server Machine**.

During the install, you will be prompted to create a root MySQL password; remember this.

You will also be able to set the MySQL server so that it runs as a service and starts automatically.

4.2. Install DeskPRO (2008)

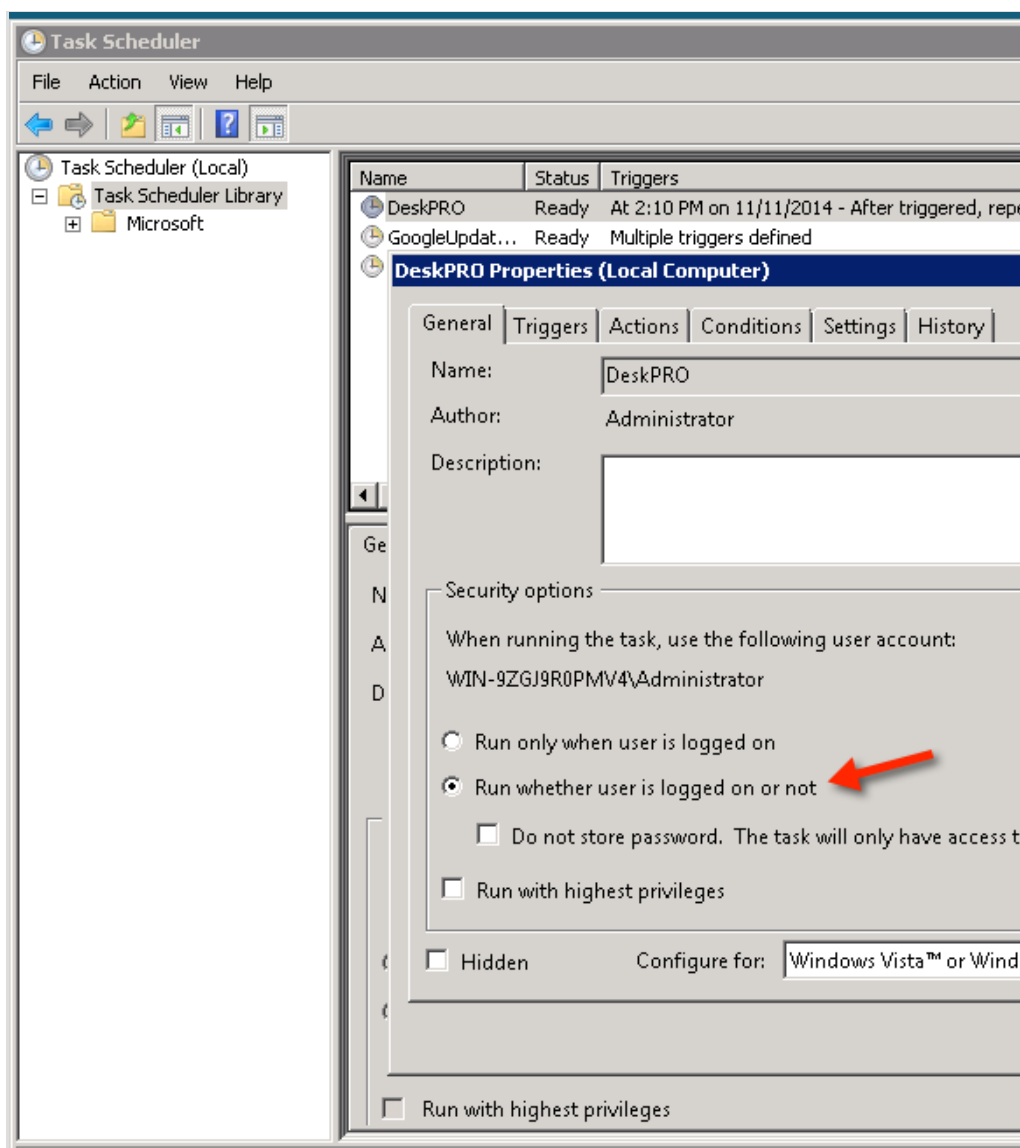
- 1 Download the latest version of DeskPRO Download from <https://www.deskpro.com/downloads/deskpro.zip>.
- 2 Find the `wwwroot` folder for IIS, e.g. `C:\inetpub\wwwroot` . You can identify this in IIS Manager by selecting **Default Web Site** in the left-hand panel, then clicking **Explore** in the right-most panel.



- 3 Remove the default files from `wwwroot` and unzip the contents of the `deskpro.zip` file.
- 4 Set up a Scheduled Task to run `php-win.exe` with `cron.php` as an argument once per minute.

You should be able to do this just by double-clicking the `schedule.bat` file in the DeskPRO install folder.

Open Task Scheduler and make sure that the DeskPRO task has been created and set it to **Run whether user is logged on or not**



If that doesn't work, see the [Setting up a Windows scheduled task](#) section below.

- 1 Make a copy of the `config.new.php` file and name it `config.php` .
- 2 Edit `config.php` :
 - Change

```
define('DP_DATABASE_HOST', 'localhost');
```

to

```
define('DP_DATABASE_HOST', '127.0.0.1');
```
 - Enter the root MySQL password in

```
define('DP_DATABASE_PASSWORD', '');
```
 - Enter your email address (for error alerts) in

```
define('DP_TECHNICAL_EMAIL', '');
```
 - In the PATHS section of `config.php`, set:
 - location of PHP CLI binary (i.e. full path to `php-win.exe` for your PHP, such as `'C:\Program Files (x86)\PHP\v5.6\php-win.exe'`)
 - location of the `mysqldump` and `mysql` binaries, such as `'C:\Program Files\MySQL\bin\mysqldump.exe'`
- 3 In your browser, go to `http://localhost` (or wherever IIS is serving the DeskPRO site), and follow the final steps.

If you get a warning that HTTP methods are being blocked by your server, go to the IIS Manager Handler Mappings panel, edit the settings for php files, click **Request Restrictions** and select **All verbs**([more details here](#)).

Please note that in production, you should follow security best practices e.g. [create a non-root mySQL user](#)

5. Installing on Windows Server 2012/IIS 8

This section describes how to quickly install DeskPRO on Windows Server 2012.

5.1. Install server software (2012)

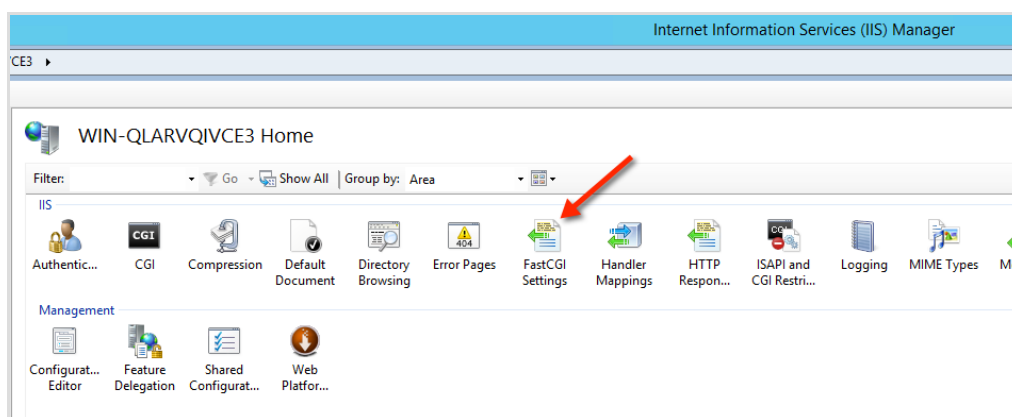
- 1 Using Server Manager, install IIS 8 and IIS Manager.
 - Go to **Manage > Add Roles and Features**
 - Select **Role-based or feature-based installation**
 - In **Server Selection**, leave the default (local) server selected.

- 2 Using the Microsoft Web Platform Installer, install:

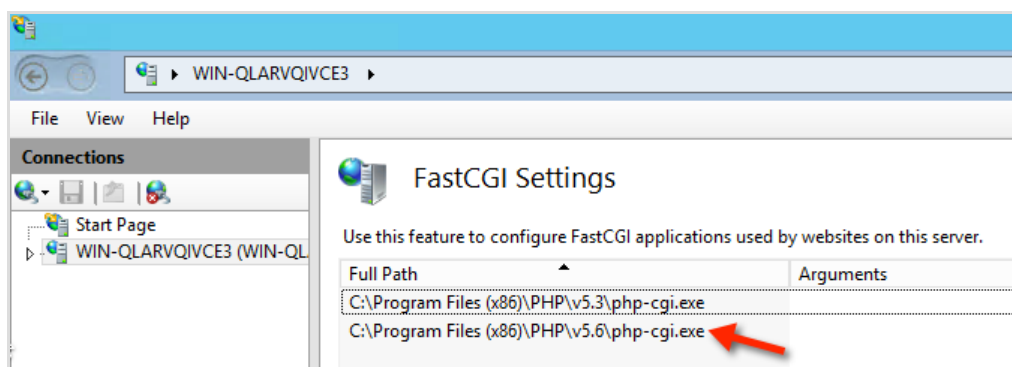
- PHP (v 5.5 recommended)
- URL Rewrite 2.0

This may download other products, including earlier versions of PHP. If you get a message about PHP Manager installation failing, don't worry.

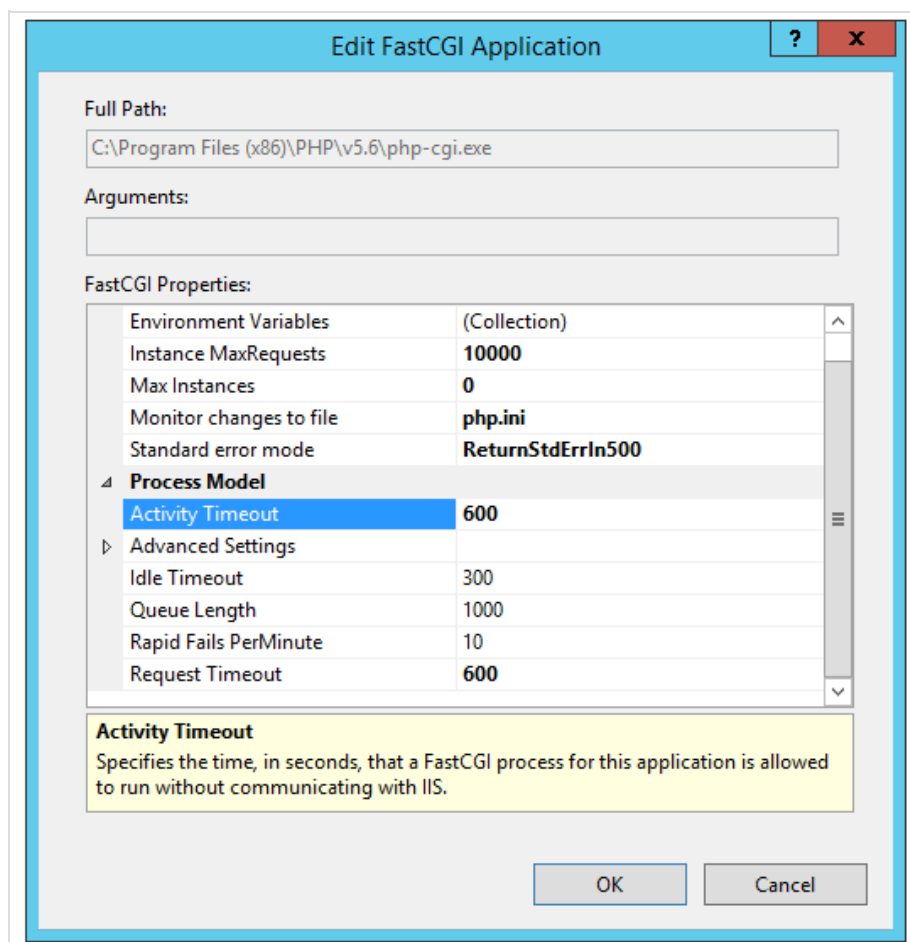
- 3 PHP should now be enabled as a FastCGI application in IIS. Look in IIS Manager and click **FastCGI Settings**.



Ensure that the version of PHP you want to use (usually, the newest one) is listed.



- If the version isn't listed, right-click on the **FastCGI Settings** pane and click **Add Application**. Enter the path to **php-cgi.exe** for the version of PHP you want to use with these settings:



- 4 Check that PHP is installed correctly by running the following from the command prompt:

```
php-cgi.exe -v
```

You should get output similar to this:

```
C:\Windows\system32>php-cgi.exe -v
PHP 5.5.11 (cgi-fcgi) (built: Apr 8 2014 15:06:59)
Copyright (c) 1997-2014 The PHP Group
Zend Engine v2.5.0, Copyright (c) 1998-2014 Zend Technologies
```

If your system has multiple versions of PHP installed, the output of this command confirms which version is actually being used.

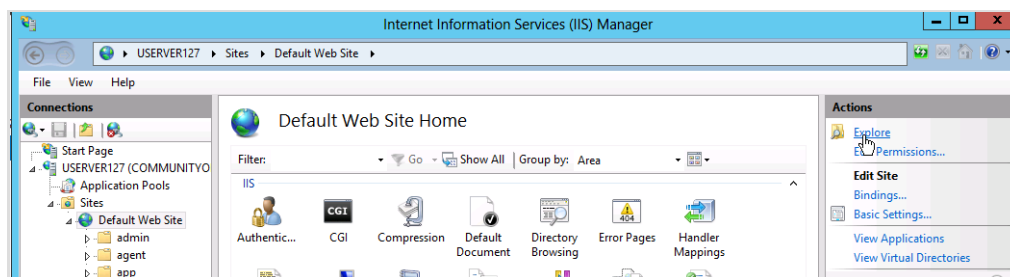
- 5 Install the latest MySQL server; an easy way to do this is using [MySQL Installer](#), select **Server Only** (despite the name, this also installs the command line client) and set the configuration type as **Server Machine**.

During the install, you will be prompted to create a root MySQL password; remember this.

You will also be able to set the MySQL server so that it runs as a service and starts automatically.

5.2. Install DeskPRO (2012)

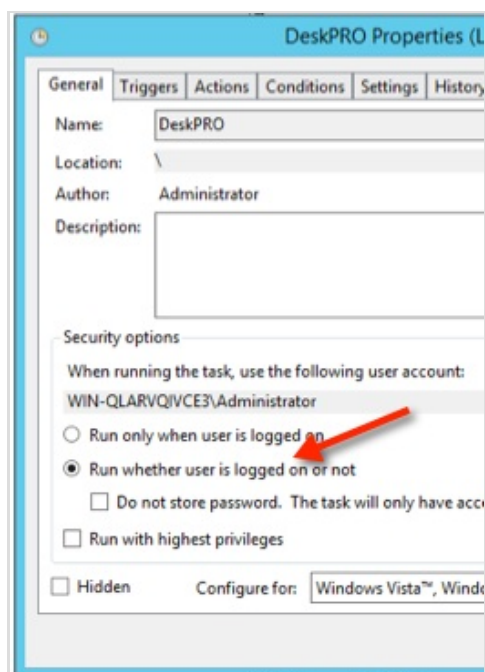
- 1 Download the latest version of DeskPRO Download from <https://www.deskpro.com/downloads/deskpro.zip>.
- 2 Find the `wwwroot` folder for IIS, e.g. `C:\inetpub\wwwroot` . You can identify this in IIS Manager by selecting **Default Web Site** in the left-hand panel, then clicking **Explore** in the right-most panel.



- 3 Remove the default files from `wwwroot` and unzip the contents of the `deskpro.zip` file.
- 4 Set up a Scheduled Task to run `php-win.exe` with `cron.php` as an argument once per minute.

You should be able to do this just by double-clicking the `schedule.bat` file in the DeskPRO install folder.

Open Task Scheduler and make sure that the DeskPRO task has been created and set it to **Run whether user is logged on or not**



If that doesn't work, see the [Setting up a Windows scheduled task](#) section below.

- 1 Make a copy of the resulting `config.new.php` file and name it `config.php`.
- 2 Edit `config.php` :
 - Change`define('DP_DATABASE_HOST', 'localhost');`to`define('DP_DATABASE_HOST', '127.0.0.1');`
 - Enter the root MySQL password in`define('DP_DATABASE_PASSWORD', '');`
 - Enter your email address (for error alerts) in`define('DP_TECHNICAL_EMAIL', '');`

- + In the PATHS section of config.php, set:
 - + location of PHP CLI binary (i.e. full path to php-win.exe for your PHP, such as 'C:\Program Files\PHP\php-win.exe')
 - + location of the mysqldump and mysql binaries, such as 'C:\Program Files\MySQL\bin\mysqldump.exe;C:\Program Files\MySQL\bin\mysql.exe'
- 3 In your browser, go to `http://localhost` (or wherever IIS is serving the DeskPRO site), and follow the final steps.

If you get a warning that HTTP methods are being blocked by your server, go to the IIS Manager Handler Mappings panel, edit the settings for php files, click **Request Restrictions** and select **All verbs**([more details here](#)).

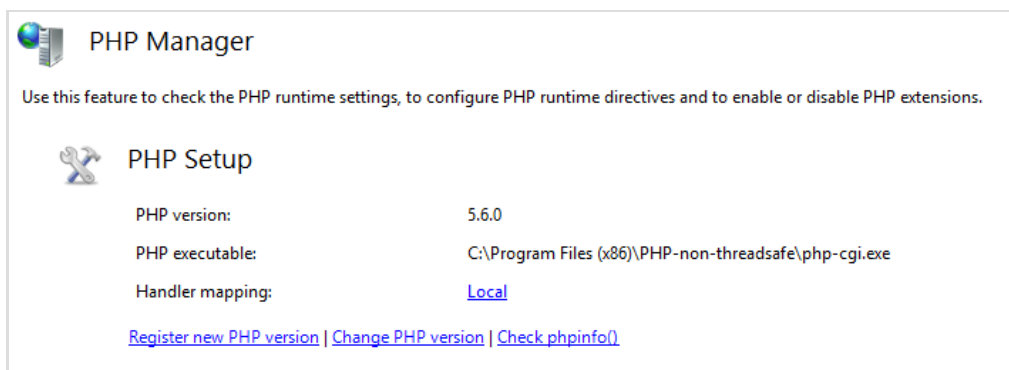
Please note that in production, you should follow security best practices e.g. [create a non-root mySQL user](#), and ensure that write access to the files in wwwroot is restricted to the IIS worker process using [Application Pool Identities](#).

6. Installing on Windows 8/IIS 8.5

This section describes how to quickly install DeskPRO using Windows 8.1/IIS Express 8.5 from scratch.

6.1. Install server software (Win 8)

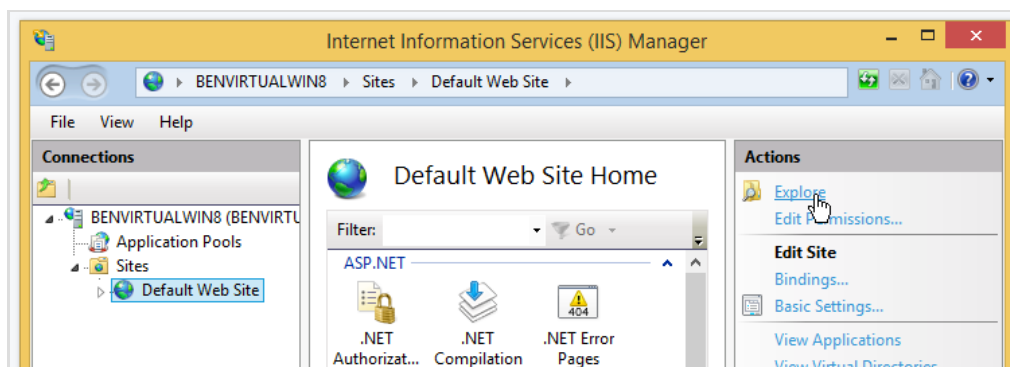
- 1 Install the [Microsoft Web Platform Installer](#).
- 2 Using the installer, find and install:
 - IIS Recommended Configuration
 - The newest available PHP for IIS Express
 - IIS Manager
 - PHP Manager
 - URL Rewrite 2.0
- 3 Follow the instructions in [the section "Enabling FastCGI support in IIS" on this page](#).
- 4 Start IIS Manager and use the PHP Manager panel. Register the version of PHP you wish to use, and use **Check phpinfo()** to ensure PHP is working correctly.



- 5 You should see a screen with the PHP version and PHP configuration information.
If you get an error, run the **php-cgi.exe** executable from the command prompt to see if there is an error message providing further information about the problem.
- 6 Install the latest MySQL server; an easy way to do this is using [MySQL Installer](#), select **Server Only** (despite the name, this also installs the command line client) and set the configuration type as **Server Machine**.
During the install, you will be prompted to create a root MySQL password; remember this.
You will also be able to set the MySQL server so that it runs as a service and starts automatically.

6.2. Install DeskPRO (Win 8)

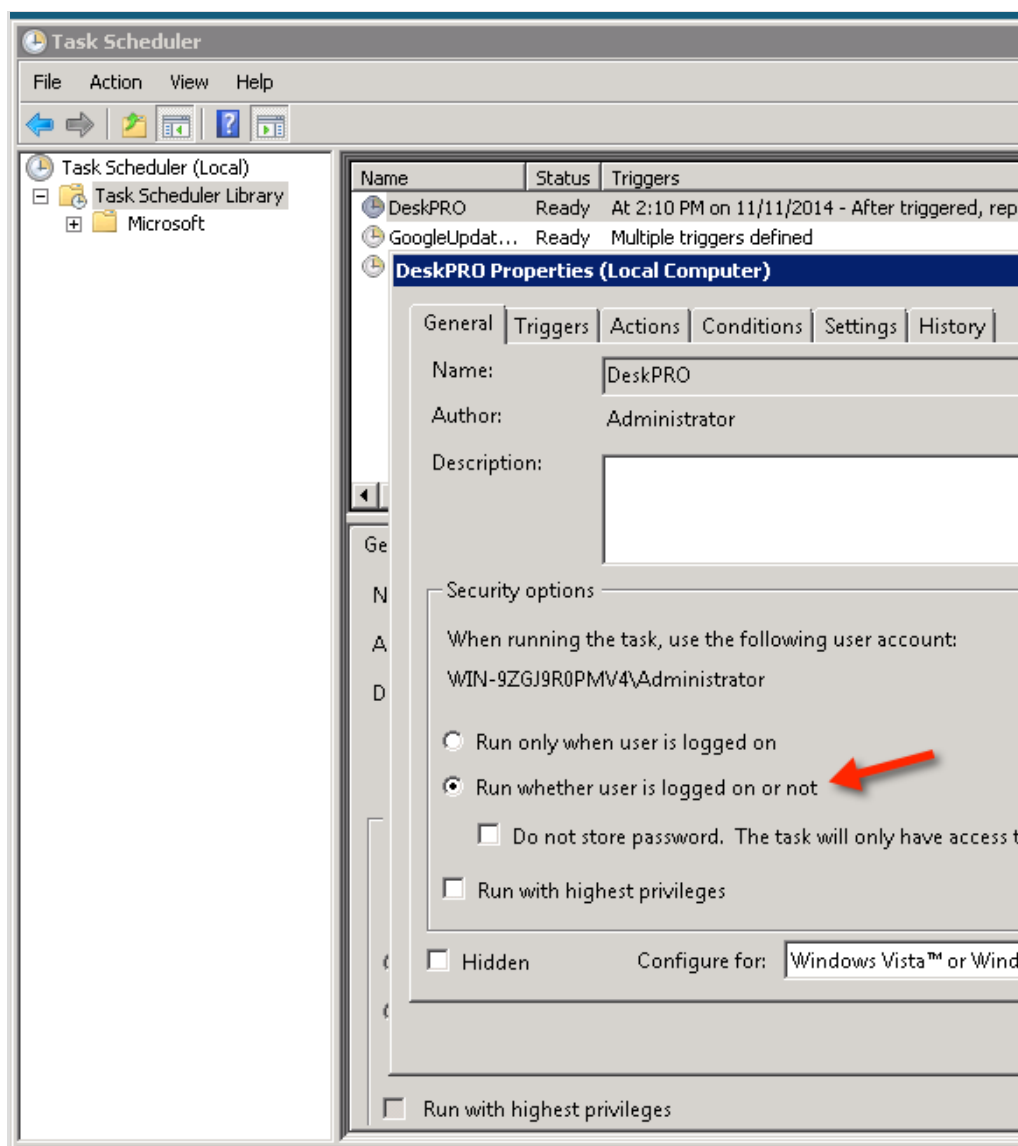
- 1 Download the latest version of DeskPRO Download from <https://www.deskpro.com/downloads/deskpro.zip>.
- 2 Find the `wwwroot` folder for IIS, e.g. `C:\inetpub\wwwroot`. You can identify this in IIS Manager by selecting **Default Web Site** in the left-hand panel, then clicking **Explore** in the right-most panel.



- 3 Remove the default files from `wwwroot` and unzip the contents of the `deskpro.zip` file.
- 4 Set up a Scheduled Task to run `php-win.exe` with `cron.php` as an argument once per minute.

You should be able to do this just by double-clicking the `schedule.bat` file in the DeskPRO install folder.

You must then find the resulting DeskPRO task in Task Scheduler and edit it to **Run whether user is logged on or not**.



If that doesn't work, see the [Setting up a Windows scheduled task](#) section below.

5 Make a copy of the `config.new.php` file and name it `config.php` .

6 Edit `config.php` :

- Change

```
define('DP_DATABASE_HOST', 'localhost');
```

to

```
define('DP_DATABASE_HOST', '127.0.0.1');
```

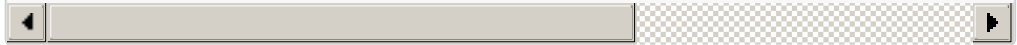
- Enter the root MySQL password in

```
define('DP_DATABASE_PASSWORD', '');
```

- Enter your email address (for error alerts) in

```
define('DP_TECHNICAL_EMAIL', '');
```

- + In the PATHS section of config.php, set:
- + location of PHP CLI binary (i.e. full path to php-win.exe for your PHP, such as 'C:\Pro
- + location of the mysqldump and mysql binaries, such as 'C:\Program Files\MySQL\bi



7

In your browser, go to <http://localhost> (or wherever IIS is serving the DeskPRO site), and follow the final steps.

If you get a warning that HTTP methods are being blocked by your server, go to the IIS Manager Handler Mappings panel, edit the settings for php files, click **Request Restrictions** and select **All verbs** ([more details here](#)).

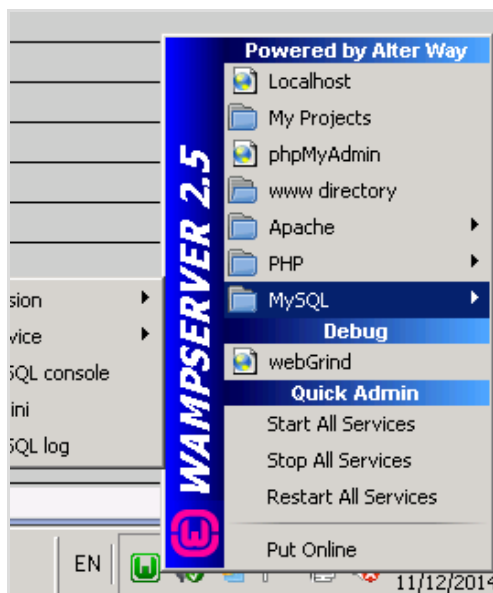
Please note that in production you should follow security best practices e.g. [create a non-root MySQL user](#) with all privileges on the 'deskpro' database and edit config.php accordingly, and ensure that write access to the files in wwwroot is restricted to the IIS worker process using [Application Pool Identities](#).

7. Installing on Windows/Apache

You can install on Windows using Apache instead of IIS. This guide was written using Windows Server 2008 and [WampServer](#).

Before you install WampServer, you should make sure you have the latest [Visual C++ Redistributable for Visual Studio 2012](#). Install the 32 bit/64 bit version, depending on whether you are going to install the 32 bit/64 bit version of WampServer.

- 1 Download and install WampServer with default settings. We recommend v2.5 with PHP5.5.
- 2 Left-click on the WampServer tray icon and click **Start All Services**.



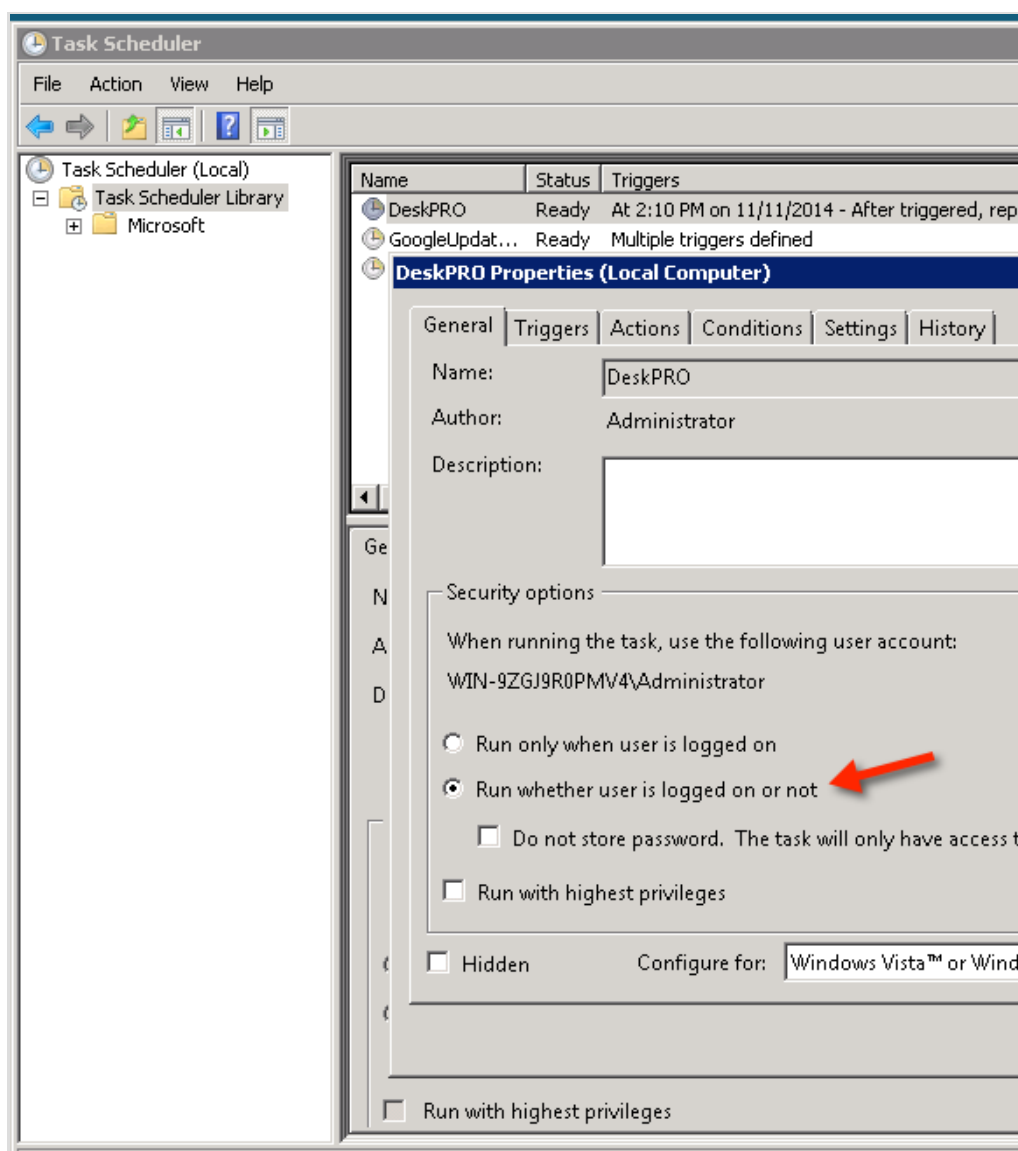
- 3 Go to `http://localhost/?phpinfo=1` to check that PHP is working correctly. Note down the location of the `php.ini` that is in use.
- 4 WampServer installs MySQL with no password for the root user. Open the tray control, select **MySQL > MySQL console**(just press Enter when prompted for a password) and set a root password:

```
SET PASSWORD FOR 'root'@'localhost' = PASSWORD('rootpass');
```

7.1. Install DeskPRO (WAMP)

- 1 Download the latest version of DeskPRO Download from <https://www.deskpro.com/downloads/deskpro.zip>.
- 2 Find the `www` folder in the WampServer install (by default, `C:\wamp\www\`).
- 3 Unzip the contents of the `deskpro.zip` file into `www`, overwriting any existing files.
- 4 Set up a Scheduled Task to run `php-win.exe` with `cron.php` as an argument once per minute.
You should be able to do this just by double-clicking the `schedule.bat` file in the DeskPRO install folder.

You must then find the resulting DeskPRO task in Task Scheduler and edit it to **Run whether user is logged on or not**.



If that doesn't work, see the [Setting up a Windows scheduled task](#) section below.

- 5 Make a copy of the `config.new.php` file and name it `config.php`.

- 6 Edit `config.php` :
 - Change`define('DP_DATABASE_HOST', 'localhost');`to`define('DP_DATABASE_HOST', '127.0.0.1');`
 - Enter the root MySQL password in`define('DP_DATABASE_PASSWORD', '');`
 - Enter your email address (for error alerts) in`define('DP_TECHNICAL_EMAIL', '');`

- + In the PATHS section of `config.php`, set:
 - + location of PHP CLI binary (i.e. full path to `php-win.exe` for your PHP, such as 'C:\Pro
 - + location of the `mysqldump` and `mysql` binaries, such as 'C:\Program Files\MySQL\bi
- 7 In your browser, go to `http://localhost` and follow the final steps.
- 8 URL rewriting is disabled in WampServer's Apache configuration by default. To enable it, go to the `httpd.conf` file (e.g. `C:\wamp\bin\apache\apache2.4.9\conf\httpd.conf`) and remove the # at the beginning of this line:

`#LoadModule rewrite_module modules/mod_rewrite.so`
- 9 The installation wizard will guide you through the final configuration steps.

Note that for production, you should follow security best practices, for example [create a non-root MySQL user](#).

8. Troubleshooting

You should be able to troubleshoot most installation problems by following the installation wizard's instructions and reading the articles it links to.

Below are instructions for resolving some issues that can prevent installation starting.

8.1. PHP configuration issues

Problem:

You are trying to change values in php.ini and they don't seem to be updating, even after you restart the webserver.

Solution:

Check what values are being used by going to `/index.php?_sys=phpinfo` .

If that doesn't work, try creating a text file containing:

```
<?php phpinfo(); ?>
```

and saving it as `info.php` in the DeskPRO install directory.

On systems where multiple versions of PHP are installed, you should make sure you are using the correct one.

On IIS, managing these issues is made easier with [PHP Manager for IIS](#). You may need to install .NET 3.5 to use it with newer versions of Windows Server.

8.2. upload_tmp_dir value won't change

Problem:

The install wizard asks you to set a new value for `upload_tmp_dir`, but the change seems to have no effect even after restarting the webserver.

Solution:

Check that there is only one `upload_tmp_dir` directive in the `php.ini` file. We've found that on some systems, there's a commented-out version near the top of the file, and a working one lower down - and the second one takes precedence.

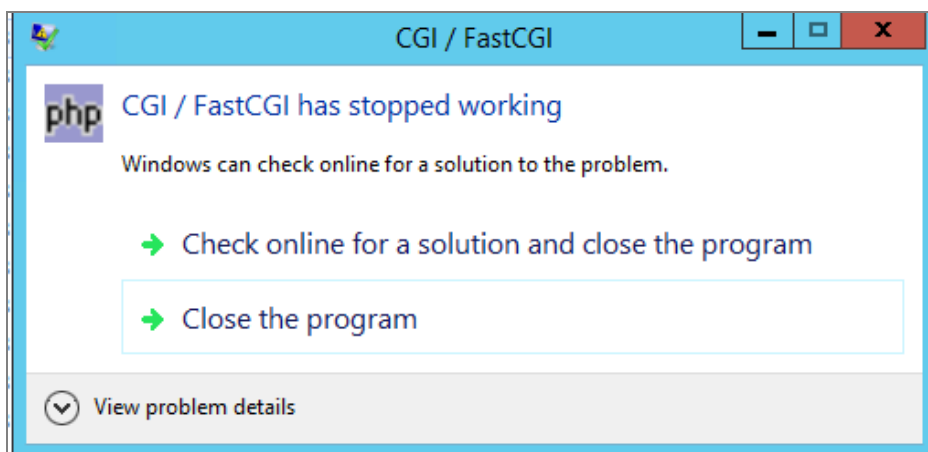
8.3. PHP/FastCGI crashes on Windows

Problem:

When you try to view your phpinfo() you may see this:

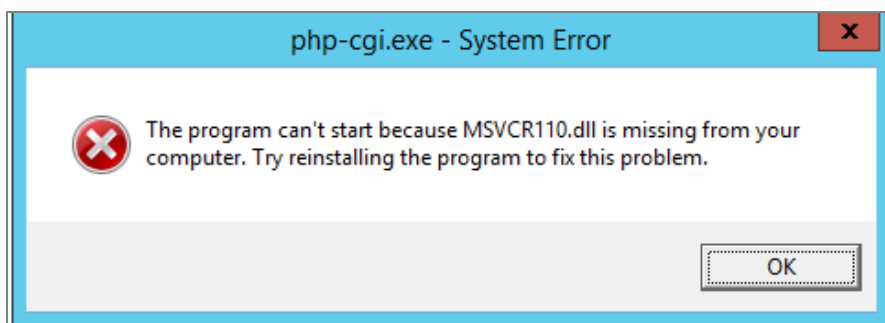
HTTP Error 500.0 - Internal Server Error

C:\PHP\php-cgi.exe - The FastCGI process exited unexpectedly



Solution 1:

Run `php-cgi.exe` from an administrator command prompt. If you get an error like this:



you need to install a Visual C++ Redistributable.

See the section VC9 and VC11 in the left-hand column of [this page](#) to find the correct Visual C++ Redistributable for your PHP version.

There are x86 and x64 versions of each Visual C++ Redistributable - you must install the version that matches your PHP, not your OS. In other words, if you are running an x64 version of Windows, but the PHP version you have is x86 (i.e. installed in C:\Program Files (x86)), install the x86 Redistributable.

Solution 2:

Another cause of this problem can be that you are running a PHP version that is incompatible with IIS. This can happen if you manually installed PHP.

You must make sure to get a **non-thread safe** version. Installing PHP through the Microsoft Web Installer should prevent this.

8.4. Existing tables were detected in your database

Problem:

The DeskPRO installer creates a 'deskpro' database within MySQL. If this is not empty (for example, you partly installed DeskPRO before, or you want to install a production version over a test helpdesk), the installation cannot proceed.

Solution:

The quickest solution is to connect to MySQL and create a new database, e.g.

```
mysql> CREATE DATABASE deskpro-live;
```

Then in `config.php`, edit this line:

```
define('DP_DATABASE_NAME', 'deskpro');
```

replacing `deskpro` with the new database name e.g. `deskpro-live`.

DeskPRO will now use the new database for installation.

9. Creating a Scheduled Task on Windows

The easiest way to configure the required Scheduled Task in Windows is to double-click on the `schedule.bat` file in your DeskPRO install folder.

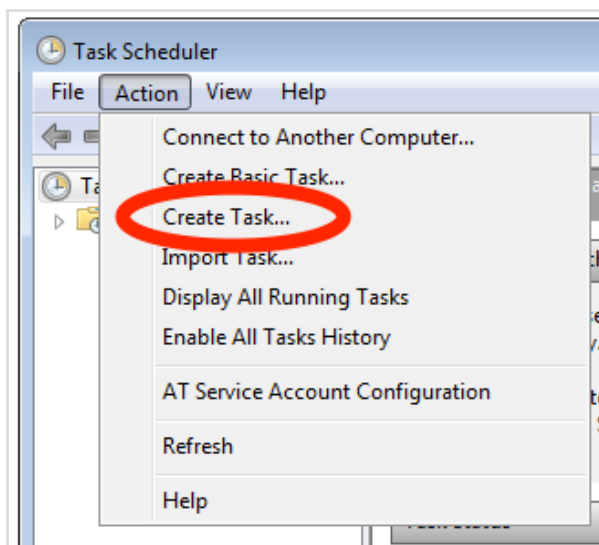
If the batch file method doesn't work, you can set up the task manually. The Scheduled Task has to run your PHP CLI executable (`php-win.exe`) with the `chron.php` file from the DeskPRO install folder as an argument, once per minute.

9.1. Open the Task Scheduler

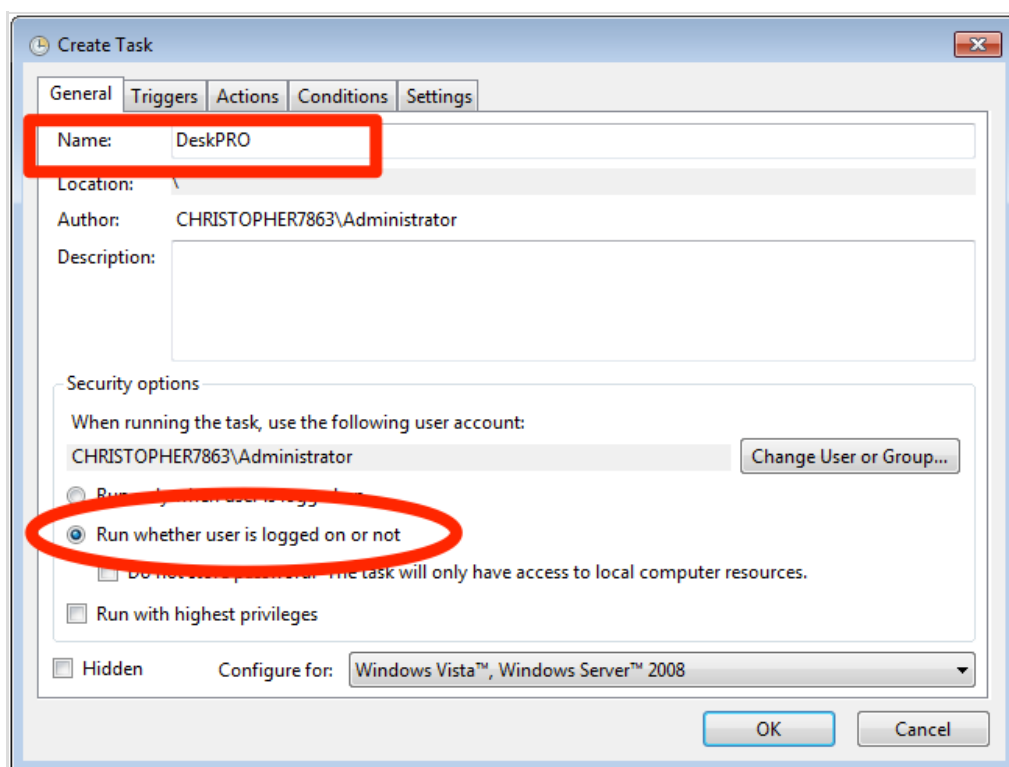
- 1 Click the **Start** button and open **Control Panel**.
- 2 Click on **System and Security** and then **Administrative Tools**.
- 3 Open **Task Scheduler**.

9.2. Create a new Task

- 1 In the left pane make sure **Task Scheduler** is selected (the first item in the list). Then from the main menu choose **Action > Create Task**

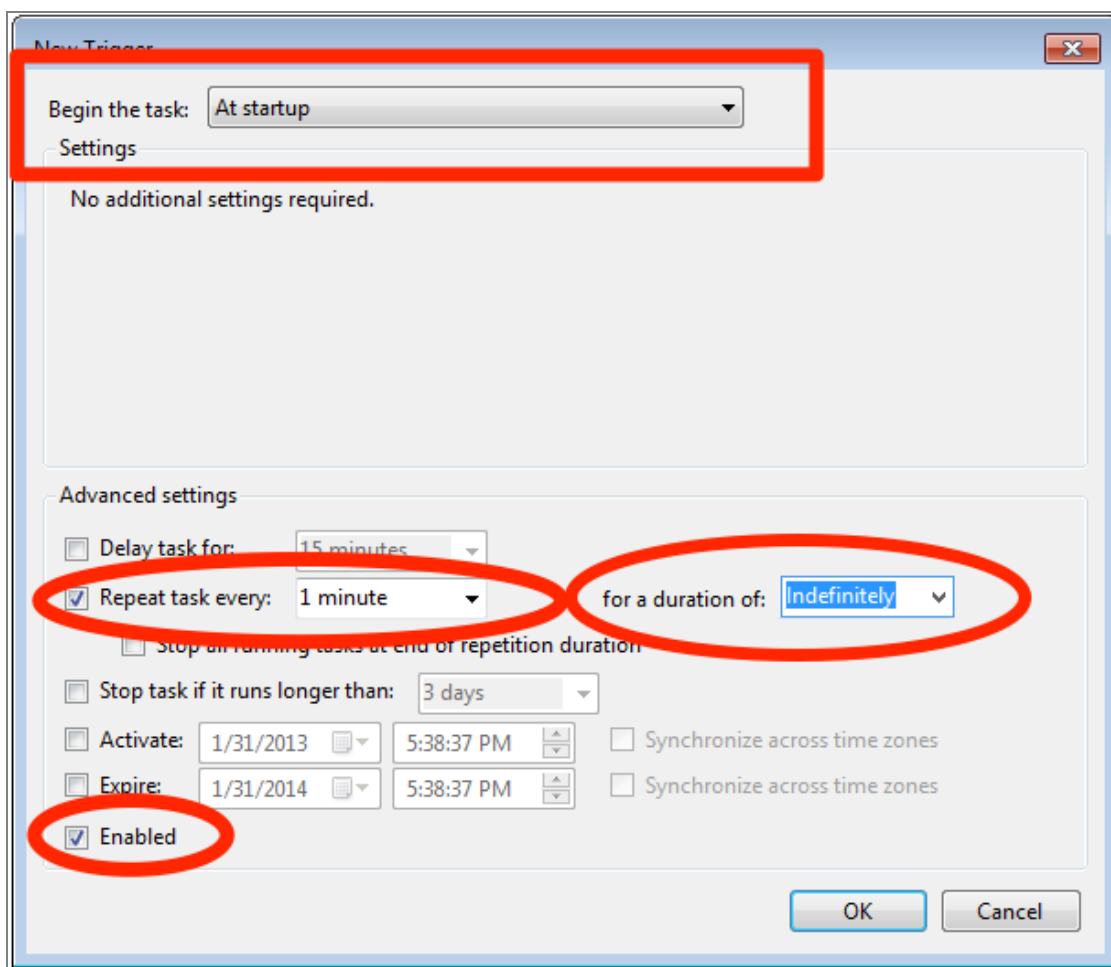


- 2 Enter a **Name** for this task (such as "DeskPRO") and then under Security Options ensure that "Run whether user is logged on or not" is selected. Note that the user that is running the scheduled task should be the same user that owns the files, or a user that has permission to write the DeskPRO files. This is required for the automatic upgrader to function. While not recommended, the most permissive user you could enter is "SYSTEM".



- 3 Switch to the "Triggers" tab at the top of window and then click the "New" button to add a new trigger.

- 4 Enter the options as displayed below. Note that for the "repeat task every" select box, the value "1 minute" is not a default option so you will need to manually type in the string "1 minute". Set the task to repeat for a duration of **Indefinitely** and click **OK**.



- 1 Switch to the "Actions" tab and click the "New" button to add a new action.
- 2 Enter the options as shown below

- The "Program/script" should be the php-win.exe executable you have installed on the system. The path may be different than what is shown here. - The "Start In" box should be the full path to the folder where you unzipped DeskPRO. The path will be different than what is shown here.

```
![(../graphics/task4.png)
```

- 1 You should now restart the computer so the task will start.

10. Create a non-root MySQL user

For simplicity, the installation steps use the root MySQL user.

For production use, you should have DeskPRO access the database through a MySQL user with more limited privileges.

You will have a MySQL client installed on your system. On Windows systems, you will have a shortcut to the command line MySQL client; on Linux/Unix, type:

```
mysql --user=root --password=rootpass deskpro
```

(replacing "rootpass" with the password for the MySQL root user you chose during installation; "deskpro" should be replaced with the name of your DeskPRO database if you are using another name).

On Windows systems you should replace localhost with 127.0.0.1 in the following steps.

Execute the following MySQL commands:

```
CREATE USER 'deskpro_user'@'localhost' IDENTIFIED BY 'password1';
```

(replacing "password1" with a more complex password of your choice).

```
GRANT ALL PRIVILEGES ON deskpro.* TO 'deskpro_user'@'localhost';
```

(where the first "deskpro" is the name of the database that DeskPRO uses).

```
FLUSH PRIVILEGES;
```

Now edit these lines of `config.php` to use the new MySQL user.

```
// The database username
define('DP_DATABASE_USER', '');

// The password for the database user
define('DP_DATABASE_PASSWORD', '');
```

11. Installing Elasticsearch

You should install the [Elasticsearch](#) specialist search technology to greatly improve search within DeskPRO.

If you don't install Elasticsearch, DeskPRO will search using MySQL. Elasticsearch offers the following improvements:

- Greatly increased search scope for agents, including the full text of ticket messages, agent notes, and chat logs.
- Agents can choose to rank results by Best Match, Last Activity or Date Created.
- Improved user portal search results.
- Search for a phrase, such as "dog and cat".
- Search is faster and results ranking is more accurate.

You can run Elasticsearch on the same server as DeskPRO, or a different server, or use a hosted Elasticsearch service from a third-party provider. The underlying technology that powers Elasticsearch uses Java, so if you want to install Elasticsearch on a server, you may need to install Java first.

You may need to upgrade your helpdesk to the latest version to be able to use Elasticsearch. If your build supports Elasticsearch you will see an Elasticsearch section in **Admin > Server**.

Server Settings	About Elastic Search Elastic Search  is a specialized search server technology. DeskPRO's default search engine uses your database, MySQL, to run queries. Elastic Search provides a faster and more powerful solution. Some benefits of using Elastic Search: • More properties are searched including full ticket messages and notes. • You can search on phrases, such as "dog and cat". • Search results are returned much faster and results are more accurate. To use this feature, you will first need to configure an ElasticSearch server or use a third-party ElasticSearch service.
Elastic Search	Elastic Search ☒ Enable Elastic Search Elastic Search URL: http://localhost:9200/ Enter the URL to your Elastic Search service. Here are a few examples: http://localhost/, https://search:9200/ or http://user:pass@host:9200/. Test Settings Save Settings

11.1. Setup with a third-party Elasticsearch provider

Several companies offer hosted Elasticsearch provision, including:

- [Found](#)
- [Qbox](#)
- [Bonsai](#)
- [Searchly](#)
- [IndexDepot](#)

Note that using hosted Elasticsearch involves sending your helpdesk data to the third-party provider. You should only use a third-party service that you trust, and you should make sure that you follow their recommendations for using their service securely.

Using a third-party service will generally not return results as fast as running your own Elasticsearch cluster locally. Most hosted services let you select where your hosted cluster will be geographically located - for best performance, pick a location as near as possible to your DeskPRO Download webserver.

When you sign up for Elasticsearch service, your provider will give you an endpoint URL.

In the DeskPRO admin interface, go to **Server > Elasticsearch**, check **Enable Elasticsearch**, and enter the endpoint URL.

For the security of helpdesk data, you should make sure to use an HTTPS endpoint.

Click **Test Settings** to check that your DeskPRO installation can contact the hosted Elasticsearch service, then click **Save Settings**.

For a more detailed setup guide, see [this article on setting up Elasticsearch using Found](#).

11.2. Elasticsearch install on Linux

- 1 If you do not already have Java installed, consult your distribution's documentation for how to install and configure Java Server JRE. For compatibility, Oracle Java rather than OpenJDK is recommended.
- 2 Download Elasticsearch from [the Download section of elasticsearch.org](https://www.elastic.co/downloads). Use the DEB or RPM package or install from a repository using apt or yum, depending on which version of Linux you're using. Install or extract the files to a directory on your server.
- 3 In the resulting /config subdirectory, edit elasticsearch.yml to restrict access to your Elasticsearch node. Search for the line that mentions "network.bind_host", delete the leading # and change it to:

```
network.bind_host: localhost
```

(This assumes you are running Elasticsearch on the same server as DeskPRO. If not, consult [the Elasticsearch documentation about network settings](#)).

- 1 Go to the resulting elasticsearch folder and run the bin/elasticsearch shell script:
- 2 You should see a series of messages as the Elasticsearch node starts.
- 3 To check that Elasticsearch is running correctly, run:

```
./bin/elasticsearch
```

```
curl -X GET http://localhost:9200/
```

You should get output similar to this:

```
{
  "status" : 200,
  "name" : "Agon",
  "version" : {
    "number" : "1.3.2",
    "build_hash" : "dee17544e2f254f3f269435345d7591939aaefd12f",
    "build_timestamp" : "2014-08-13T14:29:30Z",
    "build_snapshot" : false,
    "lucene_version" : "4.9"
  },
  "tagline" : "You Know, for Search"
}
```

- 1 In the DeskPro admin interface, go to **Server > Elasticsearch**, check **Enable Elasticsearch** and enter the hostname of the service (e.g. `http://localhost` if it is running on the same server) and the port (9200 by default).
- 2 Click **Test Settings** to check that your DeskPRO installation can connect to Elasticsearch, then **Save Settings**.

Here is a more [detailed guide on how to install Elasticsearch on Ubuntu](#).

For production use, we recommend that you consult the Elasticsearch documentation for more information about [configuration](#) and [running Elasticsearch as a service on Linux](#).

11.3. Elasticsearch install on Windows

- 1 For best performance, you should install [Java Server JRE](#).
- 2 Accept the license agreement and download the Server JRE tar.gz package.
- 3 Extract the package to a folder in Program Files. If you do not have suitable software to uncompress tar.gz files, we suggest [7-zip](#). Note that after you extract the package once, you will get a tar file without an extension which you then have to uncompress a second time.
- 4 Set the JAVA_HOME environment variable.

On Windows 7 go to **Control Panel > System > Advanced system settings > Advanced**

On Windows 8 go to **Control Panel > System > Advanced system settings**.

Click **Environment variables** and set JAVA_HOME to be the full path to the /jre folder you extracted in step 3, e.g.

```
C : \Program Files (x86) \Java \server-jre-8u20-windows-x64 \jdk1.8.0_20 \jre
```

- 1 Download the ZIP version of Elasticsearch from [the Download section of elasticsearch.org](#) and extract the files to a directory.
- 2 In the resulting /config subdirectory, edit elasticsearch.yml to restrict access to your Elasticsearch node. Search for the line that mentions "network.bind_host", delete the leading # and change it to:

```
network.bind_host: 127.0.0.1
```

(This assumes you are running Elasticsearch on the same server as DeskPRO. If not, consult [the Elasticsearch documentation about network settings](#)).

- 1 From the command prompt, go to the resulting folder and run:

```
service install `` service start
```

- 1 You should see confirmation that the service has been installed and started.
- 2 To check that Elasticsearch is running correctly, go to `http://localhost:9200/` in your browser. Depending on the browser you're using you will either see output like this, or be prompted to download a JSON file with similar content:

```
{
  "status" : 200,
  "name" : "Agon",
  "version" : {
    "number" : "1.3.2",
    "build_hash" : "dee17544e2f254f3f269435345d7591939aaefd12f",
    "build_timestamp" : "2014-08-13T14:29:30Z",
    "build_snapshot" : false,
    "lucene_version" : "4.9"
  },
  "tagline" : "You Know, for Search"
}
```
- 3 In the DeskPro admin interface, go to **Server > Elasticsearch**, check **Enable Elasticsearch** and enter the hostname of the service (e.g. `http://localhost` if it is running on the same server) and the port (9200 by default).
- 4 Click **Test Settings** ****to check that your DeskPRO installation can connect to Elasticsearch, then **Save Settings.**

For production use, we recommend that you consult the Elasticsearch documentation for more information about [configuration](#) and [running Elasticsearch as a service on Windows](#).

11.4. Troubleshooting Elasticsearch

If you receive a connection error when you click **Test Settings**, ensure that there is no firewall or other security software blocking the traffic between your DeskPRO installation and the Elasticsearch provider.

Ensure that your Elasticsearch server or hosted service has enough disk space and memory to handle indexing your helpdesk, bearing in mind that the amount of data to be indexed will grow. This is how to [check cluster health](#) and load for your own server; a third-party service will usually have a console that shows you if you have enough resources available.

If Elasticsearch is not working, viewing the logs on both the **Test Settings** popup and the **Index Status** section may show error messages which explain the problem.